Streaming Sound Texture in Auditory Scenes

Jarrod M. Hicks[1,2,3] & Josh H. McDermott[1,2,3,4]

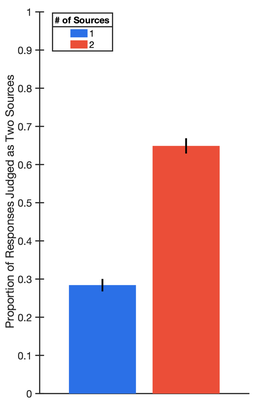
[1]Department of Brain & Cognitive Sciences (MIT), [2]McGovern Institute for Brain Research (MIT), [3]Center for Brains, Minds & Machines (MIT), [4]Program in Speech and Hearing Biosciences and Technology (Harvard)

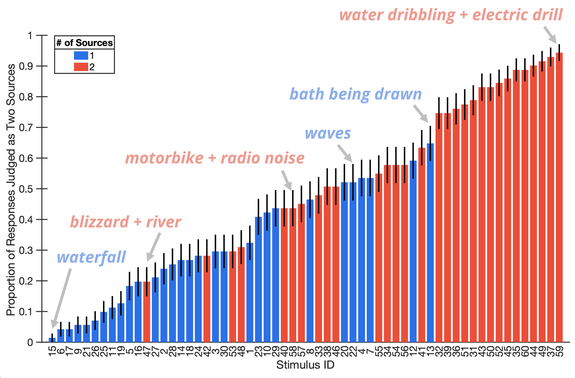
Introduction

- Sound textures are created by the superposition of similar sounds (e.g., rain falling, birds chirping, people clapping)
- Thought to be represented by statistics that summarize information over time
- Real-world auditory scenes often contain multiple concurrent textures
- Can listeners can "hear out" (i.e., stream) individual textures?
- **We characterized "texture streaming" by asking whether listeners can estimate the number of sound texture sources in an auditory scene**

Experiment 1: Streaming Judgments

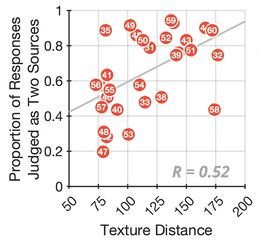
- Participants (*N=71*) heard scenes composed of 1 or 2 real-world texture recordings and judged the number of distinct sound sources
- Listeners tended to correctly judge the number of sources in each auditory scene
- Inspection of judgments for individual scenes revealed consistent patterns of errors as some combinations of textures tended to be mistakenly heard as a single stream





Oracle Model of Streaming via Similarity

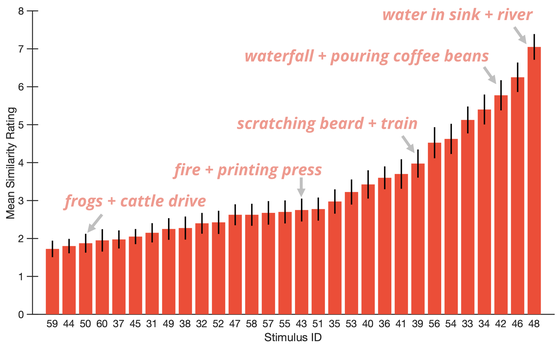
- Classical streaming is often explained via similarity of constituent elements
- If individual elements can be detected in auditory input, then grouping by similarity could determine streaming
- But it's unclear whether relevant texture features can be detected in an auditory scene, as textures are broadband and dense
- We first quantified similarity as Euclidean distance in the space of texture statistics
- We used this as an oracle model to evaluate the extent to which streaming can be predicted if textures were already separated

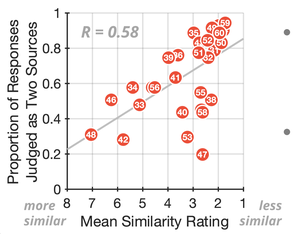


- Distance in statistics space only partially predicted streaming judgments

Experiment 2: Similarity Ratings

- Our distance metric might not completely capture texture similarity
- We collected similarity judgments (*N=40*) between texture pairs from experiment 1

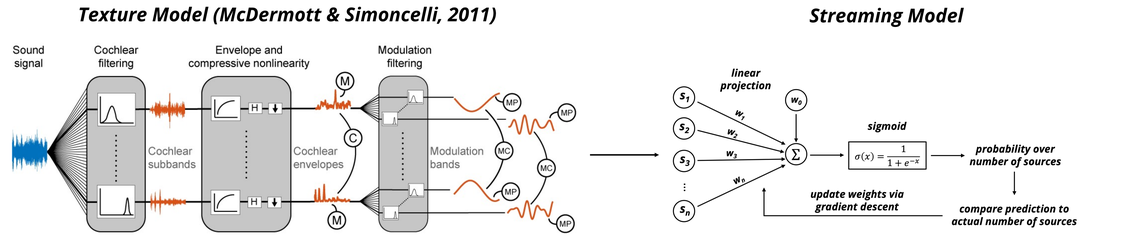


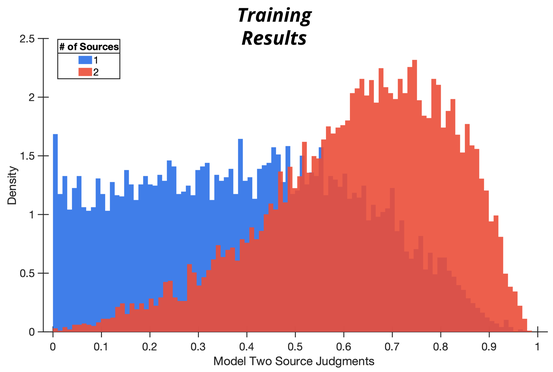


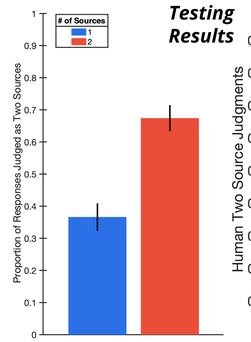
- Similarity ratings again only partially predicted streaming
- Suggests classical streaming accounts are insufficient

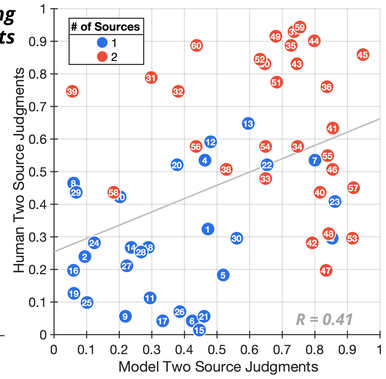
Observer Model Trained to Predict Streaming from Input Statistics

- As a test of whether streaming could be determined exclusively from input texture statistics, we trained a linear model (logistic regression) to predict the number of sources in an auditory scene
- The model measures the texture statistics of an auditory scene, performs a linear projection of the statistics and outputs a probability over the number of sources









- Model achieves human-level performance but deviates from human judgments on individual stimuli
- Suggests that a discriminative model does not account for human judgments
- Discrepancies could relate to the fact that the model does not infer the constituent streams or evaluate their plausibility

Conclusions

- We demonstrate the phenomenon of "texture streaming"—a neglected aspect of auditory scene analysis in which listeners can stream concurrent textures in auditory scenes
- Texture similarity is only partially predictive of streaming judgments → Classical account of streaming via similarity does not explain texture streaming
- A discriminative model achieves human-level performance but deviates from human judgments on individual stimuli
- Human streaming judgments may involve inference of streams and/or semantic factors not measured by texture statistics

Future Directions

- Bayesian inference model of texture streaming using a prior fit to natural sounds
- Improved (perceptually aligned) texture model using human similarity judgments

Supported by the National Science Foundation Graduate Research Fellowship under Grant No. 1122374